

Triadic Quantum Closure IV

A Typed Bridge, Gauge-Covariant Refinement, and Representation Controls

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Wings Out Research Edition ◇ formal theorem ◇ reproducible computation ◇ falsifiable physics

Abstract

TQC IV turns the previous bridge proposal into a sharper layered specification. It distinguishes descriptive roles from physical degrees of freedom, typed residual closure from scalar cancellation, classical mesh reconstruction from amplitude readout, and complex notation from an equivalent paired-real encoding. A finite bridge theorem proves that compatibility dimension is controlled by bridge structure. A triangulated prototype combines flat Regge geometry with a compact $U(1)$ connection, verifies gauge invariance to floating-point precision, and converges under mesh refinement. A coupled action landscape identifies the exact closure minimum of the current toy model. The central conclusion is bounded: the mathematics establishes a coherent bridge-and-control framework; it does not yet derive quantum mechanics or a physical Einstein–Maxwell theory.

1 The law of the document

Every claim below has a logical status. **Theorem** means proved from stated definitions. **Verification** means reproducible numerical agreement with a specified calculation. **Conjecture** means a proposed physical statement not yet established. **Control** means a matched alternative that prevents overinterpretation. **Limitation** means a known boundary of inference. **Prediction** is reserved for a held-out consequence not used to define the model.

2 Typed triadic structure

Let

$$\mathcal{I} = \{1, 2, 3\} \times \{G, E\} \times \{D, C\}, \quad |\mathcal{I}| = 12.$$

The factors index position in a triad, physics sector, and representation mode. The physical state is typed:

$$\mathcal{X} = \mathcal{X}_{GD} \times \mathcal{X}_{ED} \times \mathcal{X}_{GC} \times \mathcal{X}_{EC},$$

not automatically $\mathbb{R}^{12}$. Twelve is a **descriptive ledger**, not a degree-of-freedom claim. For mesh scale h , restriction and reconstruction are

$$R_h : \mathcal{X}_C \rightarrow \mathcal{X}_{D,h}, \quad W_h : \mathcal{X}_{D,h} \rightarrow \mathcal{X}_C.$$

The relation $R_h W_h = I$ on the discrete range while $W_h R_h = \Pi_h \neq I$ on the continuum space describes ordinary approximation. Information loss is not a probability law.

3 Closure means typed residual zero

The coupled condition is a typed residual

$$\mathcal{F}_h(X) = (R_G^D, R_E^D, \Pi_h R_G^C, \Pi_h R_E^C, B_G, B_E, B_S) = 0,$$

where $B_s = x_s^D - R_{s,h} x_s^C$ compares matching quantities through specified maps. Blocks must be scaled and gauge-compatible before their norms are combined. In general the admissible set is $\mathcal{F}_h^{-1}(\{0\})$, not a kernel. This gives the structural expression $3 + 3 = 0$ a precise meaning: six retained roles remain present while a typed residual vanishes.

4 Bridge theorem: mode selection without physics inflation

For $u, v \in \mathbb{R}^6$ and a permutation matrix P , let

$$F_P(u, v) = \begin{pmatrix} I & I \\ P & I \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}.$$

Theorem. $\ker F_P = \{(u, -u) : Pu = u\}$ and $\dim \ker F_P$ equals the number of cycles of P .

Proof. The first block requires $v = -u$. Substitution into the second gives $(P - I)u = 0$. A permutation-fixed vector is constant on each cycle, with one free scalar per cycle. $\square$

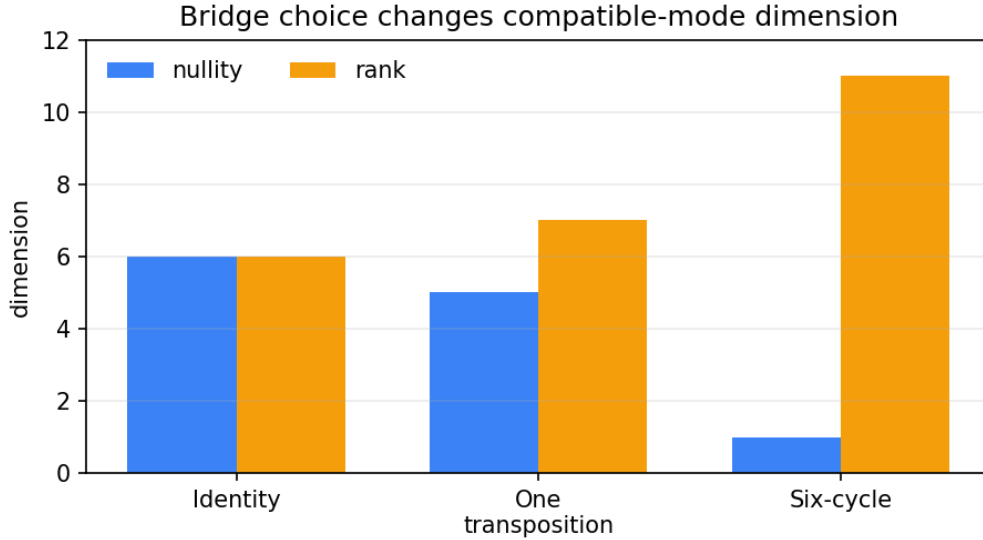


Figure 1: Bridge-controlled rank and nullity. The twelve slots do not determine the number of compatible modes.

The identity, one transposition, and six-cycle cases have nullities 6, 5, and 1. This is a classical constraint result, not a particle or quantum-state derivation.

5 Two bridges, not one

The classical bridge maps fields and meshes. The history bridge maps histories to phases and outcomes:

$$Q_\kappa(h) = e^{iS[h]/\kappa}, \quad \psi_b = \sum_{h \rightarrow b} w_h Q_\kappa(h), \quad M_b(\psi) = \frac{|\psi_b|^2}{\sum_c |\psi_c|^2}.$$

The final expression is an explicit Born-form readout, not a consequence of coarse-graining or closure. If $A : \mathbb{R} \rightarrow U(1)$ is continuous and multiplicative over additive action, then $A(S) = e^{i\alpha S}$. This conditional theorem assumes the phase group, continuity, and action additivity. Individual unit-modulus factors do not make a history kernel unitary; reversible propagation requires $K^\dagger K = I$.

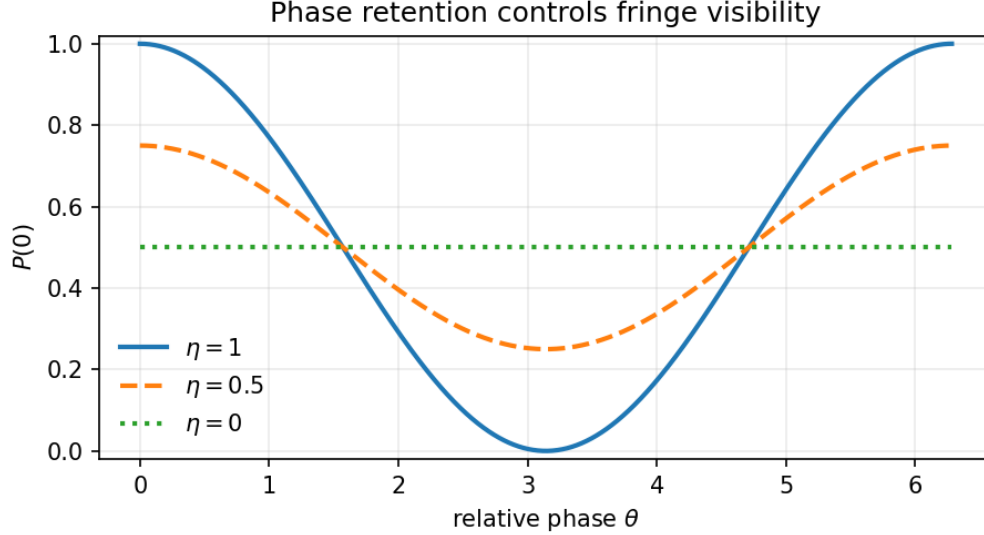


Figure 2: Explicit two-channel interference control. Coherence η is imposed; the fringes verify the chosen model rather than derive it.

5.1 The paired-real control

For $K = A + iB$, define

$$\mathcal{R}(K) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}.$$

Then $\mathcal{R}(K\psi) = \mathcal{R}(K)\mathcal{R}(\psi)$, and unitary K becomes orthogonal. A complex calculation can therefore have an exactly equivalent paired-real implementation.

6 Discrete geometric and gauge prototype

On a square domain split into counter-clockwise right triangles, assign Euclidean edge lengths ℓ_e . The two-dimensional Regge deficit at an interior vertex is

$$\epsilon_v = 2\pi - \sum_{t \ni v} \theta_{t,v}.$$

For the flat mesh, $\epsilon_v = 0$. Put a compact gauge link a_e on each oriented edge and define triangle flux by $\phi = d_1 a$. Under $a \mapsto a + d_0 \chi$, ϕ is invariant because $d_1 d_0 = 0$.

The prototype action is

$$S_h = \beta_G \sum_v \epsilon_v^2 + \beta_E \sum_t (1 - \cos \phi_t) + \lambda C_h(\epsilon, \phi),$$

with the current flat benchmark using $C_h = 0$ and $\beta_E = n^2$. The one-cell coupled diagnostic uses

$$S(d, e) = \frac{\beta_G}{2} d^2 + \beta_E (1 - \cos e) + \lambda [1 - \cos(d - e)].$$

Representation controls: what the assumptions actually select

cancellation	no	no	yes	yes
relative phase	no	no	yes	yes
unitary control	no	no	yes	yes
equivalent encoding	no	no	yes	yes
	positive real	signed real	complex phase	paired real

Figure 3: Matched representation controls. The assumptions do not uniquely privilege the word “complex.”

7 Numerical evidence

n	Wilson action	absolute error	max deficit
2	0.24967465	3.25×10^{-4}	0
4	0.24997966	2.03×10^{-5}	0
8	0.24999873	1.27×10^{-6}	0
16	0.24999992	7.95×10^{-8}	0
32	0.25000000	4.89×10^{-9}	0
64	0.25000000	$< 10^{-15}$	0

The target is $B^2/4 = 0.25$. Gauge-transformed triangle fluxes changed by at most 8.9×10^{-16} . The full project suite passes 17 tests.

Statement	Logical status
Bridge changes compatible-mode dimension	Theorem in the finite linear model
Complex phase follows from $U(1)$ character assumptions	Conditional theorem
Paired-real encoding reproduces complex propagation	Exact control
Triangle flux is gauge invariant	Discrete cochain identity
Mesh Wilson action converges in the smooth flat test	Numerical verification
Bridge derives quantum mechanics	Not established
Bridge predicts a new physical observation	Not yet tested

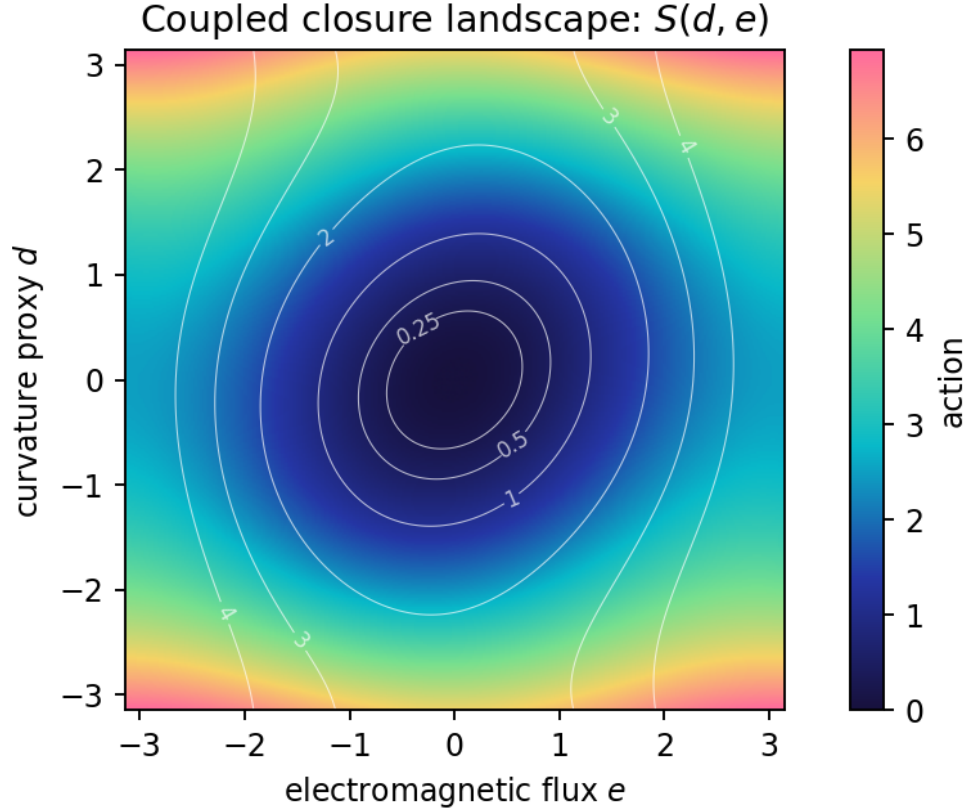


Figure 4: Fantastical-color action landscape for the coupled diagnostic. The exact minimum is at $(d, e) = (0, 0)$; color is visualization, not an added physical term.

8 What would count as a physical result?

A decisive test requires one non-flat mesh, fixed boundary data, a specified matter action, compact gauge links, a real gravitational action, and a refinement sequence. Compare positive-real, signed-real, complex-phase, and paired-real models with matched parameter counts. Measure typed residuals, gauge covariance, round-trip reconstruction error, evolution-commutation defect, unitarity error, held-out observables, runtime, and memory.

The complex model gains explanatory value only if the bridge yields a held-out restriction, compression advantage, or convergence improvement that was not built into the phase or readout definition. If paired-real and complex models agree exactly, that is a successful control: the evidence concerns a structured two-real-dimensional calculus, not a privileged notation.

9 Conclusion

TQC IV is a coherent layered program. Its **lawful core** is the separation of types, bridges, residuals, amplitudes, readout, and controls. Its present computations verify mathematics and discretization behavior. They do not yet establish that gravity and electromagnetism select quantum dynamics. That question is now narrow enough to test.

Reproducibility

Source notes and code are in the relaxation-publication-study directory. Run:

```
python3 paper/make_tqc4_figures.py
PYTHONPATH=. python3 benchmark_triangulated.py
PYTHONPATH=. python3 -m unittest discover -s tests -v
```

References

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